

DARUN MUSTAFA

Backend & AI Engineer · TypeScript · Node.js · PostgreSQL · Docker · LLM
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PROFILE

Backend & AI engineer building production-grade systems in React/TypeScript, Node.js/Fastify/Express, PostgreSQL, MongoDB and Docker — with a consistent focus on auth security, API contract design and real-time data. Built five LLM-integrated projects using Gemini, four of them live (voice-agent, darun.dev, Smart Home API, DevQuiz), including a three-provider voice pipeline with Deepgram and ElevenLabs. Experience with: multi-provider pipelines, hybrid RAG with quality scoring, Zod-validated output, token-budget enforcement and circuit breakers.

EXPERIENCE

Independent Software Engineer — Self-directed · Stockholm · Jun 2023 – Present

- Architected and shipped 8+ full-stack applications end-to-end (six highlighted below)
- Production auth from scratch: JWT + RBAC with refresh token rotation and role-scoped middleware; Argon2id and bcrypt password hashing across projects
- Enforced consistent API contracts with OpenAPI/Swagger on every backend

Personal Assistant / Night Shift Care

Stockholm · Part-time · 2018 – 2024

- Provided nightly personal care and support for a client with significant needs — coordinating care plans, schedules, and regulatory documentation reviewed by government authorities over 6 years

Earlier: Production Manager & Grocery Manager · Kurdistan Region, Iraq · 2010 – 2015

KEY PROJECTS

Research Assistant Platform · [GitHub](#) · Deployment in progress

Express · TypeScript · PostgreSQL/pgvector · Redis · Gemini API · Docker · OpenTelemetry

- Hybrid retrieval: pgvector cosine similarity + BM25 keyword search merged via Reciprocal Rank Fusion for grounded, cited answers
- Multi-node agent workflow (classify → retrieve/tool-use → evaluate) with automated quality scoring and retry logic
- 640+ Jest tests, 82% line coverage in CI; Artillery load-testing suite covering baseline, spike, and sustained-load scenarios

voice-agent · [GitHub](#) · [Live Demo](#)

Fastify 5 · TypeScript · PostgreSQL · Redis · WebSocket · Deepgram · Gemini · ElevenLabs

- Real-time voice pipeline integrating three providers: Deepgram (streaming speech-to-text), Gemini (reasoning), ElevenLabs (text-to-speech)
- Per-provider circuit breakers with graceful degradation — falls back to text input if speech-to-text fails, rather than dropping the session
- Sentry error tracking, automated accessibility testing (axe-core), dedicated latency instrumentation
- Daily token ceiling with pre-flight budget checks per Gemini/ElevenLabs call — returns 429 once exceeded, alerts at 80%

darun.dev — Portfolio Platform · [GitHub](#) · [Live Demo](#)

Fastify 5 · Prisma 7 · PostgreSQL · Redis · Gemini API · React · Tailwind · Turborepo · Cloudflare Pages

- Hybrid RAG chat grounded in my own CV and project docs, with self-maintaining embeddings driven by GitHub webhooks
- Turborepo monorepo with shared TypeScript types — Argon2id password hashing, CSRF protection, JWT auth
- Admin CMS, structured Gemini chat with Zod-validated output
- ATS source aggregation (Greenhouse, Lever, Ashby, Adzuna) with Redis caching and AI-scored matching against my CV, generating tailored pitches

CleanNation · [GitHub](#) · In progress

Bun · Fastify · Prisma · PostgreSQL (database-per-service) · Redis · Kafka

- Event-driven microservices platform — seven independent services, each with its own database
- Argon2id password hashing, JWT auth, Redis-backed rate limiting

Smart Home Automation API · [GitHub](#) · [Live Demo](#)

Node.js · TypeScript · MongoDB · MQTT · Socket.io · Gemini API · JWT/RBAC · Docker · Swagger

- Full IoT real-time loop: self-hosted MQTT broker handles device commands, Socket.io fans state to all clients in <30 ms
- Server-side Gemini integration for natural-language device commands
- Tenant-isolated RBAC at middleware layer — cross-tenant data access is structurally blocked

DevQuiz — AI quiz platform with per-user accuracy charts · [GitHub](#) · [Live Demo](#)

React · Vite · Recharts · Bun · Fastify · MongoDB · Gemini

- Gemini-generated quiz questions; React/Vite frontend with a Recharts line chart of score per attempt over the last 10 quizzes
- Bun/Fastify backend with MongoDB — Gemini output validated with Zod before it reaches the client

TECHNICAL SKILLS

Frontend React · TypeScript · Tailwind · Zustand · Axios · Vite · React Native · Expo · Framer Motion
Backend Node.js · Fastify · Express · Bun · WebSockets · MQTT · JWT/RBAC · Zod · Socket.io · OpenAPI/Swagger
AI / LLM (production) Gemini (darun.dev, voice-agent, Research Assistant, Smart Home API, DevQuiz) · Deepgram · ElevenLabs · RAG · pgvector · Agentic workflows · Prompt Engineering
AI APIs explored DeepSeek · Grok · GroqCloud
AI-assisted workflow Grok · Claude · Codex
Databases PostgreSQL · Prisma · MongoDB · Mongoose · Redis · Firebase · pgvector
Observability & Quality Sentry · OpenTelemetry · Jest · Vitest · Artillery · axe-core
DevOps Docker · GitHub Actions · Cloudflare Pages · Render · Turborepo
Security JWT/RBAC · Argon2id · bcrypt · Helmet · CSRF Protection

EDUCATION

Fullstack Developer — Open Source Track · Chas Academy · Stockholm · Sep 2025 – Jun 2027
Full-Stack Web Development Certificate · CareerFoundry (Remote) · Jun 2023 – Aug 2024
Business Administration Diploma · Choman Technical Institute · 2011 – 2013 (UHR-evaluated: SeQF Level 5)

LANGUAGES

Kurdish — Native · English — Professional · Swedish — Professional